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- 1 What I have learned after the first lab
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- 3 1. The differences between a declaration and a definition in C/C++:
- 4 - A declaration introduces a name and its type to the compiler, allowing it to recognize the name in subsequent code. It does not allocate memory or create an instance of the variable or function.
- 5 - A definition, on the other hand, allocates memory for a variable or provides the actual implementation of a function. It is where the entity is fully realized.
- 6 2. The syntax, semantics, and utility of include guards:
- 7 - Include guards are preprocessor directives used to prevent multiple inclusions of the same header file. They typically consist of `#ifndef`, `#define`, and `#endif` directives.
- 8 - The syntax is as follows:
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```\n  #ifndef HEADER_FILE_NAME_H\n  #define HEADER_FILE_NAME_H\n  // Header file content\n  #endif\n  ```
```
- 10     - The utility of include guards is to avoid redefinition errors and reduce compilation time by ensuring that the contents of a header file are only included once.
- 11 3. class (and struct) are the keywords in the C++ language that allow us to group code (member functions) and data (data members or member variables) and add new types to the C++ language;
- 12 4. The types that we add through this facility allows us to implement abstract data types (ADTs) whose operations are the member functions and whose attributes are the data members;
- 13 5. public, private, and protected are the 3 different access specifiers that allow us to restrict who is able to call our member functions and read/modify our member variables: anyone, only us and only us and deriving classes, respectily;
- 14 6. How to use getters and setters to mediate access to member variables that we have declared to be private so that we can allow users to have access to those data members in a safe, secuare manner;
- 15 7. The syntax, semantics and utility of const member functions: member functions that do not modify the state of the object on which they are called and can be called on const instances of the class;
- 16 8. The scope resolution operator (::) is used to define member functions outside the class definition and to access static members of a class.
- 17 9. The syntax, semantics and utility of declaring a derived class in C++: a derived class inherits attributes and operations from a base class, allowing for code reuse and the implementation of hierarchical relationships.
- 18 10. The syntax, semantics and utility of declaring virtual member functions in C++: virtual functions allow derived classes to override base class implementations, enabling polymorphism and dynamic binding at runtime.

24 11. The syntax required for declaring a function that takes an instance of a class as a parameter by reference and by pointer; which can call a virtual member function of that class and have the correct derived class version of that function be called (dynamic binding);

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27 // This file was referred from the part "Numismatics Are Number 1" in UChristies.md which is created by Prof. William Hawkins, and edited with more specific information by Microsoft Copilot AI.